

ORGANIC CHEMISTRY – CARBOXYLIC ACIDS

KEY VOCABULARY

Carboxylic acid: organic acid with the -COOH functional group.

Ethanoic acid: the acid in vinegar, CH_3COOH .

Weak acid: only partially ionises in water.

Salt: carboxylic acids form -anoate salts (e.g. ethanoate).

COMMON MISCONCEPTIONS

~~Carboxylic acids are strong acids.~~

✓ They are weak acids — only partly ionised.

~~Their functional group is -OH only.~~

✓ Their functional group is -COOH .

~~They don't react with carbonates.~~

✓ They fizz with carbonates, releasing CO_2 .

REACTIONS

Functional group: -COOH

Dissolve in water $\rightarrow$ weakly acidic (low $[\text{H}^+]$).

+ carbonate $\rightarrow$ salt + water + CO_2

+ metal $\rightarrow$ salt + hydrogen.

Ethanoic acid $\rightarrow$ ethanoate salts.

1

FUNCTIONAL GROUP

Give the functional group of a carboxylic acid. [1 mark]

2

WEAK ACID

Explain why carboxylic acids are weak acids. [2 marks]

3

WITH CARBONATE

Name the products of ethanoic acid + sodium carbonate. [3 marks]

4

NAME THE SALT

Name the salt made by ethanoic acid. [1 mark]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Carboxylic acids are ____ acids because they only ____ ionise.

weak

strong

partially

fully

6

COMPARE PH

Explain why ethanoic acid has a higher pH than hydrochloric acid of the same concentration. [2 marks]